

AZ-900: Microsoft Azure Fundamentals

Expert Reviewer

Overview of the AZ-900 exam

The AZ-900 exam tests your foundational knowledge of cloud services and how Microsoft Azure provides those services. Based on the provided materials, this guide focuses on the three main testing domains:

Domain	Weight	What it covers
1. Describe cloud concepts	25–30%	Cloud models, service types (IaaS/PaaS/SaaS), benefits of cloud, CapEx vs OpEx
2. Describe Azure architecture and services	35–40%	Regions/zones, resource hierarchy, compute, networking, storage
3. Describe Azure management and governance	30–35%	Security, identity, governance, cost management, support

Memory Tip

Domain 2 (architecture & services) carries the most weight — if you're short on time, prioritize compute, storage, and networking services.

Domain 1: Describe cloud concepts

This domain introduces the foundational "what" and "why" of cloud computing.

1.1 Cloud Deployment Models

- **Public cloud:** Services offered to any paying customer over the internet. The provider owns and manages the hardware, so the customer has the least control but also needs the least technical knowledge to get started. Billed pay-as-you-go (OpEx).

- **Private cloud:** A dedicated environment for a single organization (e.g., an ERP system used only inside one company). The organization has total control over resources and security, but must buy the hardware up front (CapEx). Common for legacy applications.
- **Hybrid cloud:** A mix of public and private. Considered the most flexible model because each workload can run wherever it fits best, e.g., a public-facing website on the public cloud, with sensitive data kept on a private, on-premises server.

Memory Tip

“Hybrid = best of both.” If a scenario mentions keeping something sensitive on-premises while putting something else (like a website) in the cloud, the answer is Hybrid.

1.2 Cloud Service Categories (IaaS / PaaS / SaaS)

- **Infrastructure as a Service (IaaS):** You manage the VMs, storage, and networks. It gives you the most control over the environment and is ideal for "lift-and-shift" migrations from on-premises to the cloud. Examples: Azure Virtual Machines, network interfaces.
- **Platform as a Service (PaaS):** Provides a ready-made environment for developers without the need to manage the underlying OS or servers. Examples: Visual Studio Code, Azure DevOps Services, Development frameworks.
- **Software as a Service (SaaS):** Fully built apps hosted and managed for the end user. SaaS includes everything underneath it (PaaS and IaaS). Examples: Microsoft Dynamics 365, Microsoft 365 Online.

	IaaS	PaaS	SaaS
You manage	OS, runtime, apps, data	Apps and data only	Nothing — just use the app
Provider manages	Hardware, virtualization, networking	+ OS, servers, dev tools	Everything

Best for	"Lift-and-shift" migrations; full infrastructure control	Developers who want to build apps without managing servers	End users who just need a working application
Examples	Azure Virtual Machines, network interfaces, SQL Servers deployed as VMs	Azure App Service, Azure DevOps Services, Visual Studio Code, development frameworks	Microsoft 365 Online, Microsoft Dynamics 365

Memory Tip

"I-P-S" = increasing abstraction. IaaS = Infrastructure (you do the most work). PaaS = Platform (dev tools ready-made). SaaS = Software (fully built app, zero setup).

1.3 Key Cloud Benefits and Concepts

- **High Availability:** The ability of a system to operate without downtime.
- **Scalability:** The ability to easily increase or decrease resources, like adding more virtual servers to a farm.
- **Elasticity:** The automatic scaling of resources based on demand (e.g., adding an app server automatically when CPU load hits a certain percentage).
- **Agility:** The ability to provision and de-provision resources quickly.
- **Fault Tolerance:** A system continues operating despite a component failure (e.g., a second router taking over if the first fails).
- **Reliability:** The ability to copy resources to a backup region that takes over if the primary fails.
- **Predictability:** Resources maintain a consistent behavior pattern.
- **Economies of Scale:** Gaining bulk efficiency, allowing you to get larger resources at a marginal extra cost.

Memory Tip

Elasticity vs Agility: Elasticity is automatic (the system reacts to demand on its own). Agility is about speed (how fast you, the admin, can stand something up or tear it down).

1.4 Expenditure Models: CapEx vs OpEx

- **Capital Expenditure (CapEx):** Up-front costs for physical hardware. Private clouds typically have higher CapEx because you must purchase the hardware up front.
- **Operational Expenditure (OpEx):** A consumption-based, pay-as-you-go model where users only pay for what they use, with no up-front costs. This model allows you to stop paying for resources that are no longer needed.

	CapEx (Capital Expenditure)	OpEx (Operational Expenditure)
Definition	Up-front spending on physical hardware/infrastructure	Pay-as-you-go, consumption-based spending
Payment timing	Paid before use, often in a lump sum	Paid as resources are used (though some OpEx, like reservations, can also be prepaid)
Typical cloud model	Private cloud (you must buy your own hardware)	Public cloud (Azure's default model)
Flexibility	Hard to scale down — you already own the hardware	Easy to scale down — stop paying when you stop using

Note: both models can involve up-front costs in some cases (e.g., a prepaid reserved instance is technically OpEx but paid in advance) — but the core distinction the exam tests is “buy hardware” (CapEx) vs “pay for usage” (OpEx).

Quick Recap

Models vs Services: Models = Public/Private/Hybrid (where it runs). Services = IaaS/PaaS/SaaS (how much you manage). Hybrid is the most flexible model. Public cloud runs on OpEx; private cloud requires CapEx. Elasticity is automatic scaling; Agility is rapid deployment.

Domain 1 Practice Questions

1. Which cloud model is considered the most flexible?
2. Which cloud expenditure model involves paying for resources only as they are used?
3. What term describes the automatic addition of resources to cloud infrastructure based on demand?

4. Microsoft 365 Online falls under which cloud service category?
5. Which cloud model gives organizations total control over security and resources?

Domain 1 Answer Key

1. **Hybrid** (Combines public and private for workload flexibility).
2. **OpEx / Consumption-based** (Users pay for resources used).
3. **Elasticity** (Automatic scaling).
4. **SaaS** (Software as a Service).
5. **Private** (In-house management of hardware and networking).

Domain 2: Describe Azure architecture and services

This is the heaviest-weighted domain. It covers how Azure is physically and logically organized, plus the core compute, networking, and storage services.

2.1 Physical Architecture

- **Datacenters:** Physical facilities with computing resources arranged in racks inside a climate-controlled environment.
- **Regions:** A set of datacenters connected via a low-latency network. A region can contain a single datacenter, but often contains multiple.
 - *Sovereign Regions:* Specialized instances like US DoD Central, China North, and Azure Germany (hosted by T-Systems International).
 - *Region Pairs:* If one region fails, services fail over to the other region in the pair.
- **Availability Zones:** Physically distinct datacenters with independent power, cooling, and networking. Regions that support this feature have a minimum of 3 zones.
- **Availability Sets:** Ensure instances of VMs are placed in separate update domains so that if one instance reboots for an update, the other stays running.

Feature	Protects against	Scope
Availability Set	A single datacenter's update/reboot or hardware fault	Within one datacenter
Availability Zone	An entire datacenter going offline	Across datacenters within a region (min. 3 zones)

Region Pair	An entire region going offline	Across regions
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Memory Tip

Bigger blast radius = bigger protection. Availability Set (one datacenter) < Availability Zone (multiple datacenters) < Region Pair (entire region).

2.2 Resource Organization Hierarchy

Azure organizes resources from top to bottom:

1. **Management Groups:** The top level. Can support multiple subscriptions and RBAC. They can have many children, but only *one* parent.
2. **Subscriptions:** Provide boundaries for billing and access control.
3. **Resource Groups:** Logical containers for related resources. If you delete a resource group, all resources inside are deleted. Resources moved here inherit the group's permissions. They can contain resources from multiple different regions.
4. **Resources:** The actual instances (VMs, storage, etc.).

2.3 Core Compute Services

- **Virtual Machines (VMs):** You control the OS. Charged only while running (deallocated VMs cost no compute). Provisioning requires size (CPU/RAM), storage, and networking.
- **VM Scale Sets:** Multiple identical VMs managed by a load balancer.
- **Azure Functions:** Serverless compute for event-driven code.
- **Azure App Service:** Used to host web apps, API apps, and mobile apps. Serverless capabilities are available.
- **Containers:** Run apps independent of an OS; they do not need a virtual machine to run.
- **Virtual Desktop:** Allows employees to use the exact same OS configuration regardless of the device they are on.

Memory Tip

“Serverless” in AZ-900 mainly means Azure Functions and Azure App Service, you don't provision or manage the underlying server.

2.4 Core Networking Services

- **Azure Virtual Network (VNet):** Allows VMs to communicate securely. Can be segmented into smaller, isolated "virtual subnets" (e.g., separating Sales and Marketing).
- **VNet Peering:** Connects separate VNets so they can communicate as if they were on the same network.
- **VPN Gateway:** Sends encrypted traffic from an Azure VNet to an on-premises network.
- **ExpressRoute:** A private connection to Microsoft's datacenters that does not travel over the public internet.
- **Azure DNS:** Provides domain name services, including custom domains for private VNets.
- **Load Balancers:** E.g., Azure Application Gateway (Layer 7 HTTP/S load balancer with a Web Application Firewall).
- **Endpoints:** *Public* endpoints are accessible globally, while *Private* endpoints belong to a VNet's address space.

Memory Tip

VPN Gateway = encrypted, but still over the public internet. ExpressRoute = private connection, never touches the public internet.

2.5 Core Storage Services

- **Azure Blob Storage:** Best for storing unstructured data like videos and binary files.
- **Azure File Storage:** Manages file shares in the cloud (supports SMB and NFS).
- **Azure Table Storage:** Stores structured, non-relational (NoSQL) data.
- **Azure Queue Storage:** Provides messaging between applications.

- **Azure Disks:** Block-level storage for VMs. Cost is dictated by Quality (SSD vs HDD) and Size.

Storage Access Tiers:

- **Hot:** Frequent access (e.g., daily reports).
- **Cool:** Infrequent access, stored for at least 30 days (e.g., bi-monthly reports).
- **Cold:** Rare access, stored for at least 90 days (e.g., inventory updated 2-3 times a year).
- **Archive:** Very rare access, stored for at least 180 days (e.g., annual updates).

Storage account types worth knowing: Standard general-purpose v2 (default, used for most blobs/files/queues/tables), Premium block blobs (high transaction rates), Premium file shares (support SMB and NFS), and Premium page blobs (high-performance scenarios).

Storage Access Tiers

Tier	Access pattern	Minimum storage duration	Example
Hot	Frequent access	—	Daily activity reports
Cool	Infrequent access	30 days	Reports updated every 2 months
Cold	Rare access	90 days	Inventory updated 2–3x a year
Archive	Very rare access	180 days	Annual site updates

Storage Redundancy

Option	What it does	Durability
LRS (Locally Redundant)	Copies data within a single datacenter. Lowest cost, least durable.	11 nines
ZRS (Zone Redundant)	Copies data synchronously across availability zones in the same region.	12 nines
GRS (Geo-Redundant)	Copies data 3x synchronously in the primary region, then asynchronously to a secondary region.	16 nines

GZRS (Geo-Zone Redundant)	Copies data across 3 zones in the primary region, then asynchronously to a secondary region.	Highest — combines ZRS + geo-replication
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Memory Tip

More “geo” = more nines. LRS (one place) < ZRS (zones) < GRS/GZRS (zones + another region).

2.6 Data Migration and Other Services

- **Azure Data Box:** Physical device for transferring large amounts of data (e.g., over 40 TBs) to Azure.
- **Azure Migrate:** Helps move an entire computing environment from on-premises to the cloud.
- **AzCopy:** Command-line tool to copy blobs/files.
- **Azure Storage Explorer:** GUI app to manage files and blobs.
- **Azure File Sync:** Installed locally to manage file transfers to/from Azure.
- **IoT:** Internet of Things (controlling physical/home devices remotely).
- **Azure Arc:** Manages non-Azure resources (e.g., on-premises SQL Server, Kubernetes).

Quick Recap

Physical scope: Region > Availability Zone (min. 3) > Datacenter. Logical hierarchy: Management Groups > Subscriptions > Resource Groups > Resources. For compute: VMs = full OS control, Functions = serverless event-driven code, Containers = OS-independent apps. For storage: Blobs = unstructured media, Tables = NoSQL, and redundancy options scale up as LRS → ZRS → GRS → GZRS.

Domain 2 Practice Questions

1. What is the minimum number of Availability Zones in a region that supports them?
2. Which storage tier is designed for data stored for at least 180 days and rarely accessed?
3. Which Azure service is a Layer 7 load balancer specifically designed for web applications?
4. If a resource group is deleted, what happens to the resources inside it?
5. What physical tool is used to transfer massive amounts of data (e.g., 40 TB) to Azure?

Domain 2 Answer Key

1. 3.
2. Archive.
3. Azure Application Gateway.
4. They are deleted.
5. Azure Data Box.

Domain 3: Describe Azure management and governance

This domain focuses on keeping your cloud environment secure, compliant, well-monitored, and cost-effective.

3.1 Security Models

- **Shared Responsibility Model:** Identifies who is responsible for what. The provider (Azure) is *always* responsible for the physical network, physical hosts, and the datacenter. The customer is *always* responsible for access management, accounts, and data.
- **Defense in Depth:** Security implemented in layers.
 - *Perimeter:* Filters large-scale attacks like DDoS.
 - *Network:* Limits communication among resources.
 - *Identity and Access:* Controls access to infrastructure.
 - *Physical:* Secures physical hardware.
- **Zero Trust Model:** Assumes a breach is happening at the outset. Access is granted based on authentication, not physical location. It highly utilizes "Just-In-Time" access.

3.2 Authentication vs Authorization

	Authentication (AuthN)	Authorization (AuthZ)
Answers	Who are you?	What are you allowed to do?
Examples	Username + password, biometrics (fingerprint, face/retina scan)	Granting access to a resource, role assignments
Azure tool	Microsoft Entra ID (identity verification)	RBAC (role-based permissions)

- **Multi-Factor Authentication (MFA):** Must use two *different* forms of authentication (e.g., what you know + what you are = password + fingerprint). Two of the same type (like fingerprint + retina) do not qualify as true MFA.
- **Conditional Access:** Grants/denies access based on signals such as Identity, Device, and Location.

Memory Tip

AuthN comes before AuthZ alphabetically and logically — you have to prove who you are (Authenticate) before the system decides what you can do (Authorize).

3.3 Identity and Access Tools

- **Microsoft Entra ID (Azure AD):** Provides identity services like Authentication, Single Sign-On (SSO), device management, and B2B/B2C credentialing.
 - **Azure AD Connect:** Synchronizes on-premises AD with Azure AD.
 - **Entra Domain Services:** Manages domain services like Group Policy in the cloud.
 - **B2B collaboration:** credentials external partners to access your resources.
 - **Azure AD B2C:** identity and access management for custom-built, customer-facing apps.
- **Role-Based Access Control (RBAC):** Grants access to resources based on assigned roles (e.g., giving an IT group "Contributor" status).

3.4 Policy vs RBAC

	Azure Policy	RBAC
Controls	What can be deployed/configured (rules)	Who can do what (roles)
Example use	"Only allow VMs in these regions" / "Only allow VMs up to a certain size"	"Give the IT group Contributor access to this resource group"
Format	Written in JSON	Assigned through role definitions
Grouping	A group of policies = an Initiative	—

Memory Tip

Policy = "what's allowed to exist." RBAC = "who's allowed to act."

3.5 Threat Protection

- **Azure Firewall:** Perimeter security that filters traffic by port and protocol.
- **Azure DDoS Protection:** Basic is always on for free; Standard is a paid tier.
- **Network Security Groups (NSGs):** Uses inbound/outbound rules to filter traffic to resources like subnets.
- **Microsoft Defender for Cloud (formerly Azure Security Center):** Posture management and threat protection. The Free tier offers basic assessments ; the Standard tier offers continuous monitoring.
- **Key Vault:** Securely stores secrets, passwords, encryption keys, and self-signed certificates.

3.6 Governance and Compliance

- **Azure Policy:** Enforce rules for your resources (e.g., restricting which regions resources can be built in). Policies are written in JSON. An **Initiative** is a group of policies.

- **Azure Blueprints:** Sets of standards/templates deployed across multiple subscriptions for compliance.
- **Azure Locks:** Prevent accidental changes.
 - *CanNotDelete (Delete):* Resource cannot be deleted.
 - *ReadOnly:* Resource can be read but not modified or deleted.
- **Tags:** Name/value pairs applied to resources to group them, heavily used for departmental billing.
- **Microsoft Purview:** Provides risk, compliance, and unified data governance.
- **Compliance Tools:**
 - *Service Trust Portal:* Website hosting audit reports (ISO, GDPR, FedRAMP).
 - *Trust Center:* Info on standards and privacy regulations.

Compliance terms that come up: GDPR applies to companies doing business in the EU and governs lawful disclosure of personal data; ISO (e.g., ISO 27018) addresses customer PII; NIST uses a Cybersecurity Framework (CSF) to define standards.

3.7 Management and Monitoring Tools

- **Azure Resource Manager (ARM):** The layer used to create, update, and delete resources. ARM Templates use JSON (or Bicep) and can contain parameters.
- **Interfaces:** Azure Portal (GUI) , Azure CLI (Bash scripts, Windows/macOS/Linux) , PowerShell (cmdlets). Cloud Shell runs both CLI and PowerShell.

Azure Advisor vs Monitor vs Service Health

Tool	What it does	Use it when...
Azure Advisor	Gives personalized recommendations for cost, security, reliability, and performance	You want to know how to improve your setup
Azure Monitor	Collects telemetry (like Performance Monitor); includes Log Analytics for querying trends, Alerts for thresholds (e.g., 75% RAM usage), and Application Insights for web apps	You want to track metrics, logs, or get alerted on thresholds
Azure Service Health	Tracks outages, incidents, and planned maintenance across Azure services and regions	Something seems broken and you want to check if it's a Microsoft-wide issue first

Memory Tip

Advisor = advice. **Monitor** = metrics/logs/alerts. **Service Health** = “is it just me, or is Azure down?” — always check **Service Health** first when investigating a suspected outage.

3.8 Cost Management (Pricing, Cost, and Support)

Discount and Pricing Options

Option	How it works	Best for
Spot Instances	Steepest discount; uses unused Azure capacity but can be reclaimed (decommissioned) at any time	Test/experimental VMs that can be interrupted
Reserved Instances / Capacity	Up to ~72% off in exchange for a 1–3 year commitment	Predictable, long-running workloads
Azure Hybrid Benefit	Reuse existing on-premises licenses in Azure	Organizations migrating licensed workloads (e.g., Windows Server, SQL Server)

- **Cost Factors:** Resource type, location, and size. *Note:* Outbound (egress) data transfers differ in price by region. Virtual networks themselves do not cost money.

Calculators and Tools

Tool	Purpose
Pricing Calculator	Estimates the monthly cost of an Azure configuration
TCO Calculator	Compares the total cost of an on-premises setup vs. the equivalent Azure setup. First step: define the workload.
Cost Management	Tracks actual resource costs, creates budgets, and sets budget alerts

Memory Tip

Pricing Calculator = “estimate what Azure will cost.” **TCO Calculator** = “compare Azure vs. staying on-premises.”

3.9 Support, SLAs, and Subscriptions

Support Plan	Key feature
Basic / Developer	For non-production use; Developer is business-hours support only
Standard	24x7 phone and email support
Professional Direct	Includes onboarding and training
Premier	Includes support for some non-Microsoft products

- **Service Level Agreements (SLAs):** Differ by subscription tier and service. Primarily dictate "Uptime".
- **RPO vs RTO (general IT concepts referenced in Azure SLAs):** Recovery Point Objective (RPO) is the maximum acceptable amount of data loss, measured in time. Recovery Time Objective (RTO) is how long it takes to restore service.
- **Subscription types:** Enterprise Agreement (annual contract with a committed spend), CSP — Cloud Solution Provider (subscription and solutions obtained through a Microsoft partner), and the Free account (popular services free for 12 months, plus a credit for the first 30 days).
- **Lifecycles:** Private Preview (invite-only) → Public Preview (open to everyone, generally no SLA) → General Availability (GA, fully supported).

Quick Recap

The Provider always owns physical hardware; you always own your data and accounts. Entra ID handles Identity (AuthN); RBAC handles Roles (AuthZ). Use Policy to enforce rules, Locks to prevent deletion, and Tags to organize for billing. Use the TCO calculator to compare cloud vs. on-premises cost, and the Pricing calculator to estimate a monthly Azure bill.

Domain 3 Practice Questions

1. Which Azure pricing model provides the largest discount for virtual machines that can be decommissioned at any time?
2. Which website provides Microsoft's audit reports and compliance information (like GDPR/ISO)?
3. What is the name of a collection of Azure policies grouped together?
4. Under the Shared Responsibility Model, what is the cloud provider *always* responsible for?
5. Which Azure tool analyzes your infrastructure and makes recommendations for cost, security, reliability, and performance?

Domain 3 Answer Key

1. **Spot Pricing / Spot Instances.**
2. **Service Trust Portal.**
3. **An Initiative.**
4. **The physical network / physical datacenters.**
5. **Azure Advisor.**

Final last-minute reviewer

Review these highly testable concepts right before you walk into the exam:

- **CapEx vs OpEx:** CapEx = up-front hardware costs (private cloud). OpEx = pay-as-you-go (public cloud).
- **IaaS vs PaaS vs SaaS:** IaaS = you manage OS/infra. PaaS = dev tools, Microsoft manages OS. SaaS = fully built app (e.g., M365).
- **Elasticity vs Agility:** Elasticity = automatic scaling. Agility = rapid manual deployment.
- **Region > Zone > Datacenter:** Regions with zones have a minimum of 3 zones. Each zone is one or more distinct datacenters.
- **Resource Groups:** Deleting a resource group deletes everything inside it; moving a resource into one applies that group's permissions.
- **TCO vs Pricing Calculator:** TCO = compare Azure to on-premises. Pricing = estimate your monthly Azure bill.
- **Spot vs Reserved vs Hybrid Benefit:** Spot = cheapest, interruptible. Reserved = 1–3 yr commitment, big discount. Hybrid Benefit = reuse existing on-prem licenses.

- **Advisor vs Monitor vs Service Health:** Advisor = recommendations. Monitor = telemetry/logs/alerts. Service Health = outages & maintenance.
- **Shared Responsibility:** You always own data and accounts. Azure always owns physical infrastructure.
- **Defense in Depth order:** Perimeter (DDoS) → Network → Identity & Access → Physical.
- **Policy vs RBAC:** Policy = rules on what can be deployed. RBAC = roles on who can act.
- **Authentication vs Authorization:** AuthN = proving who you are. AuthZ = what you're allowed to do.
- **Key Vault:** Stores secrets, encryption keys, and self-signed certificates.
- **Storage Redundancy order:** LRS (11 nines) < ZRS (12 nines) < GRS/GZRS (highest, replicate to another region).
- **Storage Access Tiers:** Hot (frequent) → Cool (30+ days) → Cold (90+ days) → Archive (180+ days, rarest access).
- **ExpressRoute vs VPN Gateway:** ExpressRoute = private connection, bypasses public internet. VPN Gateway = encrypted, but over the public internet.
- **VM billing:** VMs are billed only while running; a deallocated VM costs no compute.
- **Management hierarchy:** Management Groups > Subscriptions > Resource Groups > Resources.
- **DDoS Protection tiers:** Basic = always on, free. Standard = paid, enhanced mitigation.

Practice test

1. An administrator hosts Active Directory on-premises but hosts a database in the cloud because the on-premises server can't handle it. Which cloud model is this?
2. Which cloud service type provides a development environment without managing the underlying operating system?

3. Which Azure networking service carries data to Microsoft's datacenters without using the public internet?
4. An administrator needs to restrict which regions people can deploy resources into. What should they set?
5. Which Azure storage redundancy option offers 16 nines of durability by copying data to a secondary geographic region?
6. A user logs into Azure with a username and password. Which identity process just occurred?
7. What happens to a resource's permissions when it's moved into a different resource group?
8. What file format is used to build Azure Resource Manager (ARM) templates?
9. A developer wants the cheapest possible VM for an experiment that can be shut down anytime. What pricing option should they choose?
10. Which two factors does Azure use to price storage disks?
11. A business wants total control over its own hardware and security for a single-organization application. Which cloud model fits best?
12. Which Azure tool should an administrator check first when investigating a suspected Azure-wide outage?
13. A school needs to isolate network traffic for each classroom within one virtual network. What should the administrator create?
14. Which two authentication factors together would count as true MFA: a password + fingerprint, or a fingerprint + retina scan?
15. What is the minimum subscription tier required to use Microsoft Defender for Cloud (Azure Security Center)?

Answer key

- **1. Hybrid cloud** — Combines an on-premises (private) component with a public cloud component.
- **2. Platform as a Service (PaaS)** — PaaS abstracts away the OS/infrastructure and provides ready-made development tools.

- **3. ExpressRoute** — A private connection that bypasses the public internet entirely.
- **4. Azure Policy** — Policies enforce rules on resource deployment, such as allowed regions.
- **5. Geo-Redundant Storage (GRS)** — GRS replicates synchronously in the primary region and asynchronously to a secondary region, reaching 16 nines of durability.
- **6. Authentication** — Entering credentials to prove identity is authentication, not authorization.
- **7. It inherits the new resource group's permissions** — Permissions follow the resource group a resource currently sits in.
- **8. JSON** — ARM templates are built in JSON (Bicep is also supported as a higher-level language that compiles to JSON).
- **9. Spot pricing / Spot Instances** — Spot offers the steepest discount for interruptible workloads.
- **10. Quality (SSD vs HDD) and size** — These are the two factors that determine disk pricing.
- **11. Private cloud** — Private clouds give organizations total control over resources and security, at the cost of higher CapEx.
- **12. Azure Service Health** — Service Health shows real-time outage and maintenance status across Azure before you escalate further.
- **13. Virtual subnets** — Subnets divide a single VNet into smaller, isolated network segments.
- **14. Password + fingerprint** — This pairs “what you know” with “what you are” — two different factor types. Fingerprint + retina are both “what you are,” so they don't qualify as true MFA.
- **15. Free** — Microsoft Defender for Cloud's Free tier provides basic security assessments; Standard is the paid, continuous-monitoring tier.